

Pavan Raj R

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PROFESSIONAL SUMMARY

AI & ML Engineer with hands-on experience in building real-time intelligent systems including NLP-based AI Assistants, Computer Vision-powered facial recognition, automation frameworks, and data-driven applications. Strong expertise in Python, Machine Learning, Deep Learning, Data Processing, and API-driven development. Passionate about solving real-world problems through AI and deploying scalable ML models.

EDUCATION

- **Master of Computer Applications (MCA)** – AI, ML & Data Science (Pursuing)
S-VYASA Deemed to be University, Bangalore – 2024-2026
- **Bachelor of Science (BSc)** – Physics, Computer Science
Surana Autonomous College, Bangalore – 2022–2024
CGPA: **7.88**
- **12th (PUC)** – Vijaya Composite PU College – 79.5% (2019–2021)
- **10th (SSLC)** – Vivekananda International Public School – 90.88% (2018–2019)

PROJECTS

1. AI Voice Assistant using NLP, Speech Recognition & Google Gemini API

Tech: Python, NLP, SpeechRecognition, PyQt5, Gemini API, Multithreading

- Designed and developed a real-time AI voice assistant with continuous listening and intelligent command execution.
- Implemented NLP pipelines and integrated Gemini API for smart responses.
- Built a PyQt5 custom UI with non-blocking multithreaded architecture.
- Added system automation: screenshots, window control, screen lock, file operations.
- Modular structure for GUI, audio processing, and AI logic.

2. Real-Time Face Recognition & Deep Learning-Based Emotion Classification

Tech: Python, OpenCV, DeepFace, LBPH, NumPy

- Built a real-time computer vision system using webcam feed for face recognition.
- Implemented LBPH for identity detection & DeepFace for emotion classification (happy, sad, angry, neutral).
- Enabled dynamic face registration without retraining full dataset.
- Developed dedicated training (train.py) and inference (main.py) modules.

3. API-Based Quiz Application using Python, OOP & Tkinter (Open Trivia DB)

Tech: Python, Tkinter, Requests API, OOP Concepts

- Developed a GUI-based quiz game pulling questions from Open Trivia DB API.
- Used OOP classes for clean architecture and modularity.
- Integrated randomization for shuffled answer patterns.
- Designed a simple, user-friendly interface with Tkinter widgets.

4. Web Automation Testing Framework using Selenium WebDriver & Python

Tech: Selenium, Python, PyCharm

- Built automation test scripts for Orange HRM (open-source HR platform).
- Performed automated login tests, dashboard interaction, and form validations.
- Used Selenium WebDriver for DOM interactions and Python for scripting.
- Designed reusable functions for better maintainability and debugging.

TECHNICAL SKILLS

Programming: Python, Java, OOP

AI & Machine Learning: Scikit-Learn, Pandas, NumPy, Deep Learning, Feature Engineering, Model Training, Classification, Regression

Frameworks / Libraries: TensorFlow, PyTorch, OpenCV, DeepFace, Matplotlib, Seaborn

Tools: Git, PyCharm, Selenium, Jupyter Notebook

Databases: MySQL, SQL Queries

SOFT SKILLS

• Problem Solving • Critical Thinking • Teamwork • Adaptability • Communication